THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 11: **World of Algorithms** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Using $\gcd(212, 84) = 4$, the LCM of 212 and 84 is: 1

(a) 4448 **(b)** 4452 **(c)** 4456 **(d)** 4664

2. To test whether 1427 is prime, we check divisibility by 2, 3, 4, … up to $\lfloor\sqrt{1427}\rfloor$. The last number checked is: 1

(a) 37 **(b)** 38 **(c)** 713 **(d)** 36

3. A counter starts at 0. For i = 1, 2, …, 60, the counter increases by 1 whenever i is divisible by 3 or by 5. The final counter value is: 1

(a) 28 **(b)** 30 **(c)** 27 **(d)** 29

4. By Euclid's algorithm, $\gcd(2500, 1450)$ is: 1

(a) 49 **(b)** 51 **(c)** 52 **(d)** 50

5. Start with $n = 38$. Repeat: if n is even replace it by $n/2$, otherwise by $3n + 1$. The number of steps needed to reach 1 is: 1

(a) 23 **(b)** 21 **(c)** 20 **(d)** 22

6. Using Euclid's subtraction algorithm for $\gcd(1962, 1044)$, the number of subtraction steps needed is: 1

(a) 13 **(b)** 14 **(c)** 16 **(d)** 15

7. Using Euclid's subtraction algorithm for $\gcd(1353, 561)$, the number of subtraction steps needed is: 1

(a) 11 **(b)** 8 **(c)** 10 **(d)** 9

8. Set r = 0. While n > 0: r = 10r + (n mod 10), then n = quotient of n by 10. Starting with n = 8759, the final r is: 1

(a) 9668 **(b)** 9587 **(c)** 9578 **(d)** 9569

9. **Assertion (A):** In the column addition $478 + 369$, a carry is generated in the ones column. 1
Reason (R): $8 + 9 = 17$, which is greater than 9.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** $24 \bmod 6 = 0$. 1
Reason (R): $0 \bmod 6 = 0$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the gcd of 287 and 245 by Euclid's subtraction algorithm and use it to find their LCM. 2

12. Find the gcd of 128, 192 and 144 by applying Euclid's subtraction algorithm twice: first to two numbers and then to the result and the third. 2

13. A floor of 128 cm by 164 cm is to be covered by identical square tiles, as large as possible, with no cutting. Use Euclid's subtraction algorithm to find the tile side and the number of tiles. 2

14. Find the gcd of 399 and 308 by Euclid's subtraction algorithm and use it to find their LCM. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Trace: count = 0; for i = 1 to 8: for j = i to 8: count = count + 1. Find the final count and explain the pattern. 3

16. Trace: total = 0; for i = 1 to 7: if i is a multiple of 4, add i^2 to total. List the numbers added and the final total. 3

17. Start with n = 30. Repeat: if n is even replace it by n/2, otherwise by 3n + 1, until n = 1. Write the sequence and the number of steps. 3

OR

Trace: count = 0; for i = 1 to 9: for j = i to 9: count = count + 1. Find the final count and explain the pattern. 3

18. Trace: total = 0; for i = 1 to 10: if i is a multiple of 3, add i^2 to total. List the numbers added and the final total. 3

OR

Start with n = 17. Repeat: if n is even replace it by n/2, otherwise by 3n + 1, until n = 1. Write the sequence and the number of steps. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Weekdays repeat after every 7 days, so Rohan uses the remainder after division by 7 (the mod operation) to find the day of the week. Number the days Sunday $= 0$, Monday $= 1$, Tuesday $= 2$, Wednesday $= 3$, Thursday $= 4$, Friday $= 5$, Saturday $= 6$. 5

(i) Find $29 \bmod 7$.

(ii) Find $100 \bmod 7$.

(iii) Today is Monday. What day will it be after 100 days?

(iv) Today is Thursday. What day will it be after 45 days?

20. Find the largest number that divides both 553 and 385 exactly, using Euclid's subtraction algorithm. Hence list all the common divisors of the two numbers. 5

OR

Find the gcd of 430, 258 and 222 by applying Euclid's subtraction algorithm twice: first to two numbers and then to the result and the third. 5

END OF QUESTION PAPER — Total: 40 Marks